

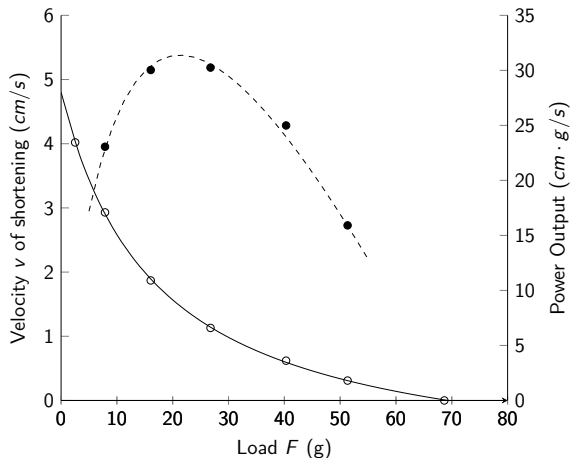
Optimization Problems

Getting the most out of a situation is something one often strives for. This is especially the case when considering complex systems that contain objectives and constraints whose relationship with each other have unclear tradeoffs and implications. Such systems arise in many applications in sciences, business and everyday life.

- ▶ What levels of light and/or nutrients yield the best crops?
- ▶ What dosage/frequency of a drug will effectively fight the disease without doing harm to the patient?
- ▶ Which parameters make a given function come closest to some measured values?

Example: Muscle contraction

muscles shorten against (not too heavy) loads



A. V. Hill (1938): $(F + a)(v + b) = d$, $a = 16.936\text{g}$, $b = 1.173 \frac{\text{cm}}{\text{s}}$,
 $d = 101.179 \frac{\text{g} \cdot \text{cm}}{\text{s}}$

Question: What is the maximal power (rate of doing work, “effect”) of this muscle? Power: $P = F \cdot v$

Maxima and Minima (Sections 20.1 and 20.2)

- ▶ We say that $f(x)$ has a local maximum/minimum at $x=a$ if $f(x)$ is less than/greater than $f(a)$ at points surrounding $f(a)$.
- ▶ We say that $f(x)$ has a global maximum/minimum if $f(x)$ attains its largest/smallest value at $f(a)$.

Observations: Let f be differentiable on (a, b) .

- ▶ If f is increasing on some interval (a, b) , then $f' > 0$ on (a, b) .
- ▶ If f is decreasing on some interval (a, b) , then $f' < 0$ on (a, b) .

Consequences:

- ▶ If $f(c)$ is a maximum, then $f' > 0$ to the left of c and $f' < 0$ to the right of c .
- ▶ If $f(c)$ is a minimum, then $f' < 0$ to the left of c and $f' > 0$ to the right of c .
- ▶ The points at which f has a local extremum (i.e. maximum or minimum) on (a, b) satisfy $f' = 0$ if f' is continuous.

Remember: Check as well the boundary points!

Example 1.1 (20.1, Sketching a function).

Classify the critical points of $f(x) = x^4 - 4x^2 + 3$ and sketch the graph.

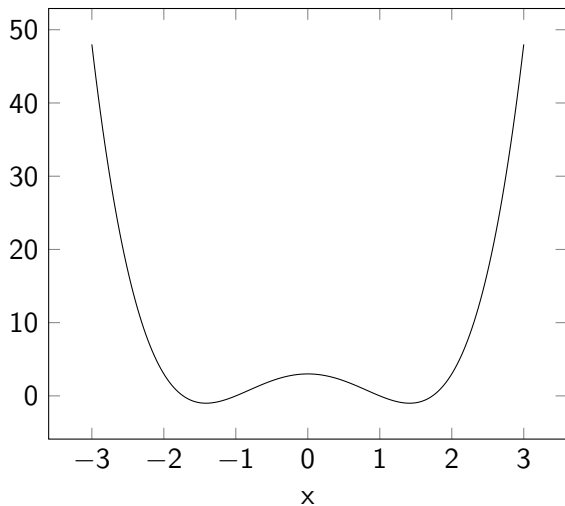
- ▶ The domain is $(-\infty, \infty)$.
- ▶ $f'(x) = 4x^3 - 8x = 4x(x^2 - 2) \implies$ critical points are $-\sqrt{2}, 0, \sqrt{2}$.
- ▶ Sign chart:

$$\begin{array}{ccccccc} - & & 0 & + & 0 & - & 0 & + & \left| \begin{array}{c} f'(x) \\ x \end{array} \right. \\ & & -\sqrt{2} & & 0 & & \sqrt{2} & & \end{array}$$

$\implies$ global minima at $x = \pm\sqrt{2}$, local maximum at $x = 0$

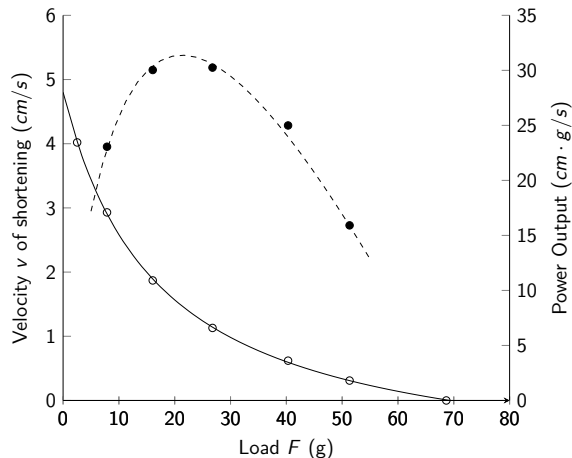
- ▶ $f(-\sqrt{2}) = f(\sqrt{2}) = -1$ and $f(0) = 3$

$$x^4 - 4x^2 + 3$$



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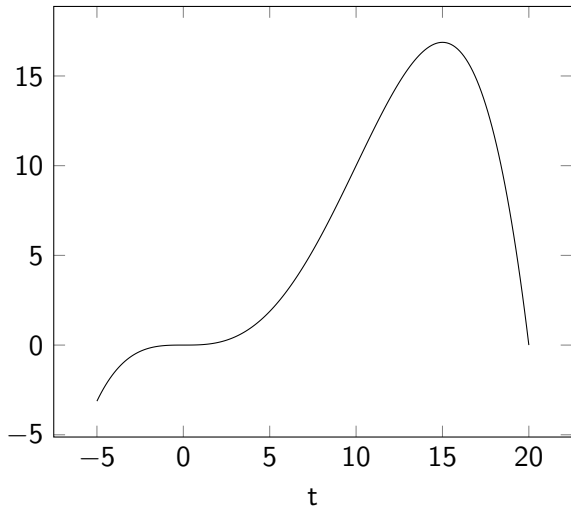
Example 1.2 (Exercise 20.19, PollEverywhere).

Epidemiologists have found a new disease, for which they estimate that t days after its first observation in the community, the percentage of population infected is approximated by

$$P(t) = \frac{20t^3 - t^4}{1000}.$$

After how many days is the percentage of the population infected at a maximum?

$$\frac{20t^3 - t^4}{1000}$$



Second derivatives test (Section 20.4)

Assume $f'(c) = 0$.

- ▶ When $f(c)$ is a maximum, then $f' > 0$ to the left of c and $f' < 0$ to the right of c . Thus f' is strictly monotonically decreasing in c . If f'' exists, it suffices therefore to test whether $f''(c) < 0$.
- ▶ When $f(c)$ is a minimum, then $f' < 0$ to the left of c and $f' > 0$ to the right of c . Thus f' is strictly monotonically increasing in c . If f'' exists, it suffices therefore to test whether $f''(c) > 0$.

Example (cmp. exercise 20.8).

Suppose the concentration $C(t)$ of a drug in the blood stream t hours after injecting it is given by

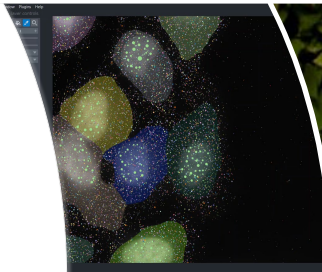
$$C(t) = \frac{6t}{2t^2 + 1}.$$

What is the highest concentration the drug will achieve in the blood stream?



Next Lecture: February Image analysis

Basic properties of digital
images and image analysis.



Jonathan Brewer

